

PHY108L: Prelab 5

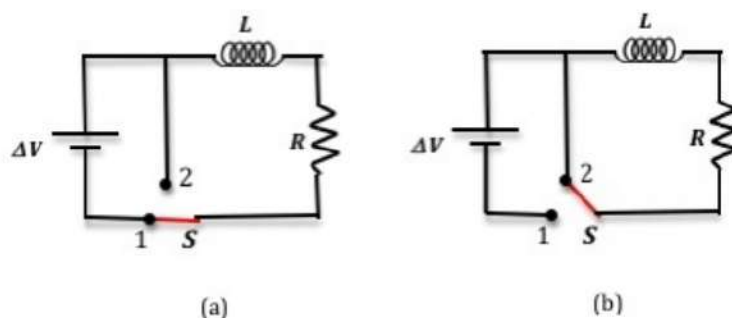
Date:

Name:

Student ID:

These questions need to be completed before starting the lab. Please show all workings.

#1 In the circuit of Figure, let $V_0=2.0\text{V}$, $R=4.0\Omega$, and $L=4.0\text{H}$. (i) what is the time constant of the circuit? (ii) What are the current in the circuit and the magnitude of the induced emf across the inductor at $t = \tau_L$, and at $t = 5.0 \tau_L$?



#2 In a RL Series circuit, a constant emf 6v is introduced where the value of resistor is $1\text{k}\Omega$ and inductor is 100mH . What will be induced inductive emf when time is $50 \mu\text{sec}$ and the time constant?